MIL-A-50210 (MU)
6 December 1968
SUPERSEDES
PA-PD-227
20 August 1952

MILITARY SPECIFICATION

ACID, MIXED
(FOR USE IN NITRATION OF EXPLOSIVES)

1. SCOPE

1.1 This Military Specification covers mixed acid consisting primarily of sulfuric acid and nitric acid in productions required for a particular nitrating process.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

JAN-N-183 - Acid, Nitric (For Ordnance Use)
JAN-A-179 - Acid, Sulfuric and Oleum

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specifications to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

FSC: 1376

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CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 49 CFR 171-179

(The Interstate Commerce Commission Rules are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402. Orders for the above publication should cite "49 CFR 171-179, (latest revision)."

3. REQUIREMENTS

3.1 Materials.-The mixed acid shall be made with sulfuric acid conforming to JAN-A-179, and nitric acid conforming to JAN-A-183.

3.2 Suspended matter.-0.05 percent, maximum.

3.3 Ash.-0.5 percent, maximum.

3.4 Acid content.-As specified by contract or purchase order.

3.5 Total sulfuric acid.-As specified by contract or purchase order.

3.6 Total nitric acid.-As specified by contract or purchase order.

3.7 Nitrosylsulfuric acid.-As specified by contract or purchase order.

3.8 Actual sulfuric acid.-As specified by contract or purchase order.

3.9 Actual nitric acid.-As specified by contract or purchase order.

3.10 Water.-As specified by contract or purchase order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves

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the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to standard MIL-STD-109 in order to define terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of batches or product inspected (shall be made available upon request by the contracting officer).

c. Results obtained, by defect code, for all inspections performed (shall be made available upon request by the contracting officer).

d. Drawing, specification number and date, together with an identification and date of changes.

e. A statement thal all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are in file and available for review.

f. Number of pounds in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing,

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not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements.

4.2.2 Sampling

4.2.2.1 Containers.-Ten percent, but in no case more than ten or fewer than three, of the containers shall be selected by the inspector so as to be representative of the lot. If there are fewer than three containers in the lot, all the containers shall be sampled. The material shall be mixed to a uniform consistency throughout and approximately 8 ounces of it shall be taken from each selected container. Each of these primary samples shall be placed in an airtight glass bottle having a ground-glass stopper and labeled so that the container from which it was taken can be identified. A composite sample of approximately 8 ounces shall be made from equal portions of the primary samples. The composite sample shall be thoroughly mixed and placed in an airtight glass bottle having a ground glass stopper and labeled to show the name of the material, manufacturer, plant, contract or purchase order number, lot number and lot size. All acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform, the inspector may require that any primary sample be tested for compliance with the requirements of the purchase description. All primary samples shall be held for possible future examination should the composite sample fail to comply with the requirements. If any sample fails to comply with the requirements specified the lot shall be rejected.

4.2.2.2 Tank cars.-By means of a clean, weighted, small-neck, glass bottle, approximately eight ounces of the material shall be removed. The unstoppered bottle shall be rinsed with the material being sampled and lowered to the bottom of the tank car. The bottle shall be immediately withdrawn to the surface. In order to obtain a representative cross-section of the material, the speed of lowering and raising the bottle shall be uniform and so regulated that the bottle is just filled as it reaches the surface of the liquid. This sample shall be placed in an air-tight glass bottle having a ground-glass stopper, and labeled to show the name of the material, manufacturer, plant, contract or purchase order number, lot number, and lot size. All acceptance tests shall be made on the composite sample representative of the lot. If any sample fails to comply with the requirements specified, the lot shall be rejected.

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4.3 Test methods and procedures

4.3.1 Suspended matter, Major defect, Code No. 01001, (see 6.2).-Transfer approximately 40 grams of the well mixed sample to a 30 ml glass-stoppered weighing bottle. Stopper the bottle and weigh. Pour most of this weighed portion from the weighing bottle through a previously tared Gooch crucible (containing a carefully prepared and dried asbestos mat) attached to a suction flask with the vacuum turned on. Reweigh the acid weighing bottle and take the weight of the sample as the difference between the two weights of the bottle. Replace the flask containing the filtrate with another flask; retain the filtered acid for the ash determination (see 4.3.2). Wash the mud or suspended material held on the asbestos mat with at least five 20 ml portions of anhydrous alcohol. The anhydrous alcohol used may be prepared by allowing 95 percent alcohol, denatured with 0.5 percent of benzene, to stand over anhydrous copper sulfate, and filtering the alcohol before use. Dry the crucible for 3 hours at 125 $\pm$ 2 degrees C., cool in a desiccator, and weigh. Calculate the increase in weight of the crucible as percentage of suspended matter.

4.3.2 Ash, Major defect, Code No. 02001.-Place the filtrate obtained from the determination of suspended matter (see 4.3.1) in a tared platinum or silica dish, heat slowly with a Bunsen flame under a hood until the evolution of fumes has ceased and then heat to a dull red for two or three minutes, cool in a desiccator, and weigh. Heat for a further short period to determine if constant weight has been obtained. Calculate the increase in weight of the dish as percentage of ash.

4.3.3 Total acidity, Major defect, Code No. 03001.-Transfer 10 ml of the sample to a tared glass-stoppered weighing bottle by means of a 10 ml pipette with the tip cut off (so that delivery can be made rapidly) and fitted with a rubber suction bulb. In draining the sample from the pipette, care should be taken so that no acid touches the ground glass portion of the weighing bottle. Stopper the bottle quickly and weigh the assembly accurately. Support a six inch glass funnel on a ring stand and fit the apex of the funnel with a rubber stopper mounted on a stout glass rod at least 6 inches long. Place the stopper assembly in the funnel outlet and fill the funnel two-thirds full with distilled water. The upper part of the glass rod attached to the stopper should project above the surface of the water. Carefully release the stopper to allow the stem of the funnel to fill with water and then restopper quickly. Place a calibrated 1-liter volumetric flask beneath the funnel with the funnel's delivery stem inside the neck of the flask.

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Hold the weighing bottle containing the weighed sample with one hand and hold the glass stopper of the bottle firmly in place with glass forceps held in the other hand. Invert the weighing bottle and lower it into the water in the funnel until the glass stopper is below the surface of the water, being careful not to touch the water with the fingers. Carefully loosen the glass stopper with the glass forceps and allow the acid to fall very slowly in a thin stream to the bottom of the funnel. Brown fumes will form in the weighing bottle. Do not allow these fumes to escape. When the acid has been emptied from the weighing bottle, release the glass stopper and permit it to fall to the bottom of the funnel, but continue to hold the weighing bottle with the neck still submerged until the brown fumes are absorbed by the water. (This can be hastened by gently swirling the weighing bottle, being careful both to keep the mouth of the bottle submerged and the fingers from touching the water). After all brown fumes are absorbed, allow the weighing bottle to empty by holding it above the level of the liquid in the funnel. Thoroughly wash the weighing bottle and cover with a stream of distilled water, allowing the washings to fall into the funnel. Loosen the rubber stopper in the apex of the funnel by carefully moving the glass rod from side to side to allow the contents of the funnel to flow into the volumetric flask. Thoroughly rinse the glass forceps and the rubber stopped assembly, catching the washings in the flask. Dilute the contents of the flask to the 1-liter mark with distilled water. Stopper the flask, mix well and allow to cool to room temperature. If any contraction of volume occurs upon cooling, adjust to the 1-liter mark by addition of more water and again mix thoroughly. Take a 100 ml aliquot measured in a calibrated aliquoting flask (calibrated to hold 100 ml) and quantitatively transfer the solution to a clean 400 ml beaker. Rinse the aliquoting flask several times with distilled water and add the rinsings to the contents of the beaker. Add two drops of 0.1% methyl orange indicator and titrate with 0.33N standard sodium hydroxide solution (using a 100 ml calibrated burette) until the color changes to an orange yellow. Add three drops of 0.1% methyl red and continue titrating dropwise until the color changes sharply from red to canary yellow. The slow color change of methyl orange serves as a warning of the approaching end point and the sharp color change of methyl red indicates the exact end point. This titration can be made by using methyl red indicator alone in cases where the approximate titration value is known. Apply any necessary temperature corrections to the final burette reading. Record the titration value and hold for the calculation of total nitric acid content.

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4.3.4 Total sulfuric acid, Major defect, Code No. 04001.- Take another 100 ml aliquot of the drowned sample prepared as described in the method for total acidity (see 4.3.3) and quantitatively transfer it to a large white porcelain evaporating dish (16 to 17 cm in diameter). Evaporate the aliquot on a steam bath with the aid of a slight current clean air to the first evidence of darkening of contents in the evaporating dish. The time required to accomplish this should be between two and two and a half hours. Rinse down the sides of the evaporating dish with 10 ml of water delivered from a pipette and evaporate as before. Repeat until free from nitric acid as shown by the vapors over the dish failing to turn moist blue litmus paper red. This second evaporation should require 20 to 30 minutes. Remove the evaporating dish from the steam bath and add 100-150 ml of distilled water. Add 3 drops of 0.1% methyl red indicator and titrate with the same 0.33N standard sodium hydroxide solution used for the total acidity determination. Apply any necessary temperature correction to the final burette reading. Calculate the percent of total sulfuric acid as follows:

$$\text{Total } H_2SO_4\% = \frac{A \times B \times 4.9038}{C}$$

Where:

A = ml of sodium hydroxide solution used in titration
B = normality of sodium hydroxide used in titration.
C = weight of sample in aliquot (1/10 of original sample)

4.3.5 Total nitric acid, Major defect, Code No. 05001.- Calculate the total nitric acid as follows:

$$\text{Total } HNO_3\% = \frac{(F - A) \times B \times 6.3016}{C}$$

Where:

F = ml of sodium hydroxide solution used in titration of total acidity
A = ml of sodium hydroxide solution used in titration of total sulfuric acid
B = normality of sodium hydroxide used in both titrations
C = weight of sample in aliquot (1/10 of original sample).

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4.3.6 Nitrosylsulfuric acid contents, Major defect, Code No. 06001.-Using the same 10 ml pipette that was used in the total acidity determination (see 4.3.3), measure as accurately as possible the same quantity of acid as was used for that determination. Deliver the sample below the surface of 150 ml of distilled water container in a 300 ml Erlenmeyer Flask. The top of the pipette should be immersed in the water to within 1/4 inch of the bottom of the flask. As the last portion of the acid runs out, gradually raise the pipette until the tip just touches the surface of the water, and allow to drain from this position for the same period of time that was allowed when transferring the sample in the total acidity determination. Titrate the contents of the flask with recently standardized 0.2N potassium permanganate solution as follows: Toward the end of the titration, add the permanganate solution dropwise, allowing the color to completely discharge between each addition. The end point is taken when the pink color produced by one drop of the permanganate solution remains visible for at least one minute. Calculate the percent nitrosylsulfuric acid as follows:

$$\text{Nitrosylsulfuric acid, \%} = \frac{V \times N \times 6.354}{W}$$

Where:

V = ml of $KMnO_4$ solution used
N = normality of $KMnO_4$ solution
W = grams of sample (see 4.3.3).

4.3.7 Actual sulfuric acid, Major defect, Code No. 07001.- Calculate the actual sulfuric acid content of the mixed acid as follows:

$$\text{Actual } H_2SO_4\% = A - (0.7718 \times B)$$

Where:

A = percent total H_2SO_4
B = percent $HNOSO_4$

4.3.8 Actual nitric acid, Major defect, Code No. 08001.- Calculate the actual nitric acid content of the mixed acid as follows:

$$\text{Actual } HNO_3\% = A - (0.4959 \times B)$$

Where:

A = percent total HNO_3
B = percent $HNOSO_4$

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4.3.9 Water, Major defect, Code No. 09001.-Calculate the water content of the mixed acid as follows:

Water, % = 100 - (A +B +C)

Where:

A = percent actual H_2SO_4
B = percent actual HNO_3
C = percent $HNOSO_4$

5. PREPARATION FOR DELIVERY

5.1 Packing.-Acid shall be packed in a manner that shall conform to the requirements contained in the Code of Federal Regulations, CFR 49, 171-179, for shipment of acid, and in a manner that will insure safe delivery from manufacturer's plant to the first receiving activity.

5.2 Marking.-Marking shall be in accordance with Standard MIL-STD-129, and shall conform to the Code of Federal Regulations, CFR 49.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

Custodian:
Army-MU

Preparing Activity:
ARMY-MU

Project Number: 1376-A024

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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